JIRAN

SYSTEM ARCHITECTURE & DESIGN

Jiran is an AI-powered platform that enables small retailers to efficiently manage excess inventory through real-time stock exchange. By leveraging intelligent demand-supply matching, the system reduces waste, improves profitability, and enhances supply chain efficiency.

PROBLEM

- Small retailers operate with isolated inventory systems, limiting visibility beyond their own stock
- Excess inventory leads to wastage and locked capital, while other stores face shortages
- Lack of demand insight results in inefficient stock planning and missed sales opportunities
- Existing systems do not support real-time, network-level inventory optimization

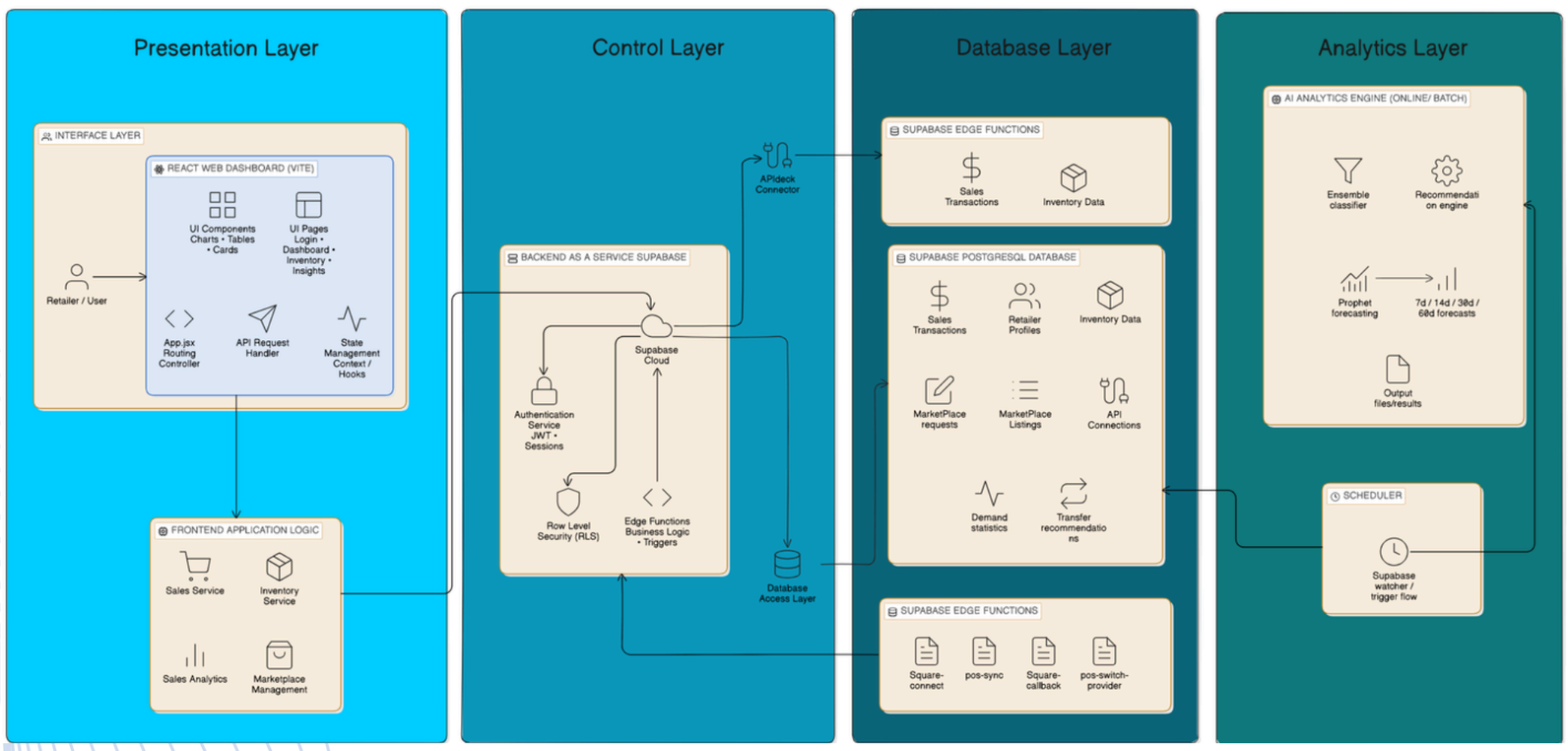
SOLUTION

- A connected inventory platform that enables real-time stock visibility across retailers
- AI-driven analysis to predict demand and classify inventory health
- Intelligent system for redistributing surplus stock within the network
- Seamless integration with existing POS systems for continuous data flow and minimal disruption

HOW IT WORKS

- Retailers upload or sync inventory data via POS or manual import
- System processes historical and real-time data for each product-store pair
- AI models forecast demand and classify stock status
- Platform identifies surplus and shortage across the retailer network
- System generates stock transfer or restock recommendations
- Retailers can list, request, buy, or sell inventory through the marketplace
- Transactions are completed and inventory is updated in real time

SYSTEM ARCHITECTURE



- Frontend (React): User interface for inventory management, analytics, and marketplace interaction
- Backend (Node.js): Handles business logic, API requests, authentication, and data processing
- Database (Supabase/PostgreSQL): Stores inventory, transactions, retailer data, and analytics
- AI Engine (Python ML API):Time-series forecasting using Prophet
- Ensemble classification model for stock status prediction
- API Layer: Connects frontend, backend, and ML services for real-time communication
- External Integrations: POS systems (e.g., Square) and geolocation services

CORE FEATURES

- AI-powered demand forecasting and stock classification
- Real-time inventory tracking and visibility
- AI-driven stock transfer recommendations between retailers
- Automated alerts for low, slow, and excess inventory
- Integrated marketplace for retailer-to-retailer transactions
- POS system integration for seamless data synchronization

TECH STACK

- **Frontend:** React.js (Vite)
- **Backend:** Supabase
- **Database:** PostgreSQL (Supabase)
- **Query Language:** SQL
- **Backend / ML Language:** Python
- **Frontend Language:** JavaScript
- **Real-Time Updates:** Supabase Realtime
- **Machine Learning:** Ensemble Models (XGBoost, Gradient Boosting, Random Forest)
- **Forecasting:** Prophet
- **Mapping:** OpenStreetMap
- **Cloud & Deployment:** Supabase Cloud, Docker
- **Version Control:** GitHub
- **ML Development:** Jupyter Notebook